NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

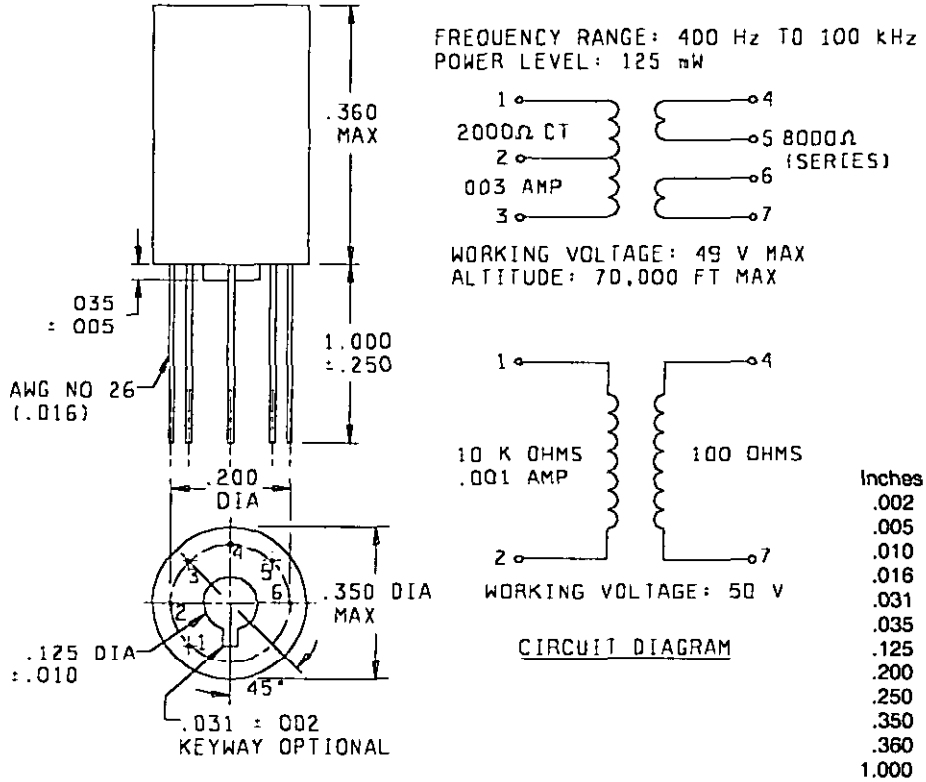
MIL-PRF-27/97B
14 January 1988
SUPERSEDING
MIL-T-27/97A
3 May 1979

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY, SUB-MINIATURE, TF5R21ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification and the latest issue of MIL-T-27.



Inches	mm
.002	0.05
.005	0.13
.010	0.25
.016	0.41
.031	0.79
.035	0.89
.125	3.18
.200	5.08
.250	6.35
.350	8.89
.360	9.14
1.000	25.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The number of terminals shall be as shown in circuit diagram.
4. Terminals are spaced the same as "TO-76" transistors and micrologic elements.
5. For vibration and shock test the specimen shall be rigidly mounted on a printed circuit board.

FIGURE 1. Dimensions and configurations.

(B) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:31Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Electrical ratings See table I
Primary power level 125 mW

Design and construction
Dimensions and configuration See figure 1.
Duty cycle Continuous.
Terminals: Pin for printed circuit
Material: Type D per MIL-STD-1276
Diameter 0.016 inch.
Length 1.000 ±0.250 inch.
Weight: 2.4 grams, maximum.
Operating temperature range. -55°C to +105°C.
Altitude 70,000 feet

Terminal strength Method 211 of MIL-STD-202, test condition A, 2 pounds.

Dielectric withstanding voltage
At sea level. 100 V rms
At reduced barometric pressure: 100 V rms

Electrical characteristics
Harmonic distortion: Total harmonic content of output 5 percent at 1 kHz
No load (ct voltage balance only): With 10 volts, 10 kHz applied to (1-3), unbalance (4-5) and (6-7) shall be 3 percent, maximum
Self-resonant frequency See table II.
Frequency response See table II
Polarity See table II.

Vibration, high frequency Method 204, MIL-STD-202, test condition B

TABLE I Electrical ratings.

Dash no.	Circuit diagram	Primary impedance	Secondary impedance	Primary dc current	Primary dc resistance ±25 percent	Secondary dc resistance ±25 percent	Working voltage
			ohms	mA	ohms	ohms	volts
(B) 01	A	2,000Ω ct	(4-5) 8,000 (6-7) 8,000	3	200	(4-(5+6)-7) 450	49
02	B	10 kΩ	100	1	1000	10	50

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:31Z
Check the source to verify that this is the current version before use.

TABLE II Electrical characteristics

Dash no	Frequency response ±3 dB at 400 Hz to 100 kHz and 1 mW			Resonance, self resonant frequency	Polarity (additive with terminals connected)
	Z_S	Z_L	E_S	(min)	
			volts	kHz	
01	2 kΩ	8 kΩ	3	40	3-4 and 5-6
02	10 kΩ	100 ohms	6.32	--	3-4

Part number. M27/97- (and dash no from table I).

TABLE III. Substitution data. 1/

Part no. M27/97-	CAGE	Manufacturer's part number
01	31669	C265
02	31669	C3375

1/ These transformers are substitutable for the manufacturer's part numbers. This in no way implies that the manufacturer's part number is substitutable for the military part number. This information is not intended to be complete.

Custodians.
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0697)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:31Z
Check the source to verify that this is the current version before use.